MINI PROJECT REPORT

Lifestyle-Based Early Detection of Sleep Disorder Risk Using Machine Learning

1. Basic Details

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Course & Sem.	B.Tech CSE(AIML) 3rd Year/6th Sem
Project Title	Lifestyle-Based Early Detection of Sleep Disorder Risk using Machine Learning

2. Problem Statement

Sleep disorders affect a significant portion of the global population and are closely linked to chronic health conditions such as cardiovascular disease, obesity, and mental health disorders. Traditional clinical diagnosis relies on expensive polysomnography (PSG) tests, making early intervention inaccessible for many.

This project aims to build a machine learning classification system that predicts the risk level of sleep disorders — Healthy, Mild, Moderate, or Severe — purely from lifestyle and physiological features such as sleep duration, stress score, caffeine intake, physical activity, and chronotype.

The objective is to enable proactive, non-invasive, and cost-effective early screening that can be deployed as a digital health tool, allowing individuals to assess their sleep disorder risk before seeking formal clinical evaluation.

3. Dataset Description

The dataset used is the Sleep Health and Daily Performance Dataset sourced from Kaggle (uploaded by Mohan Krishna Thalla). It contains 100,000 records and 32 columns capturing a comprehensive range of daily lifestyle, physiological, and environmental variables.

The target variable is sleep_disorder_risk, a four-class categorical label: Healthy (54,156), Mild (33,479), Moderate (8,299), and Severe (4,066). The dataset is complete with zero missing values across all features.

Key features include: sleep_duration_hrs, sleep_quality_score, stress_score, caffeine_mg_before_bed, screen_time_before_bed_mins, exercise_day, steps_that_day, rem_percentage, deep_sleep_percentage, and mental_health_condition.

4. Exploratory Data Analysis (EDA)

4.1 Dataset Preview

The dataset contains 100,000 rows and 32 columns. Numerical features include age, BMI, sleep duration, stress score, heart rate, and more. Categorical features include gender, occupation, country, chronotype, mental health condition, season, and day type. There are no null values in the dataset.

4.2 Histogram – Sleep Duration Distribution

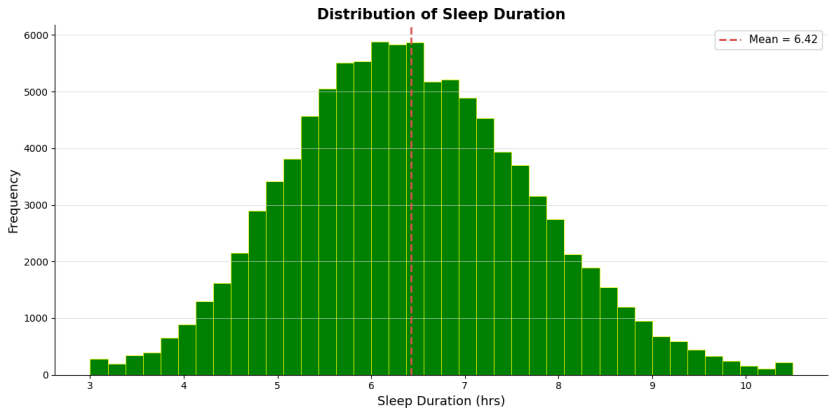


Figure 1: Distribution of Sleep Duration (hrs) across 100,000 records

Observation: Sleep duration spans approximately 3-10.5 hrs with a mean of 6.42 hrs. The distribution is roughly bell-shaped with a slight left skew, indicating a sub-population sleeping well below the recommended 7-9 hrs - a key marker for sleep disorder risk.

4.3 Class Distribution – Sleep Disorder Risk

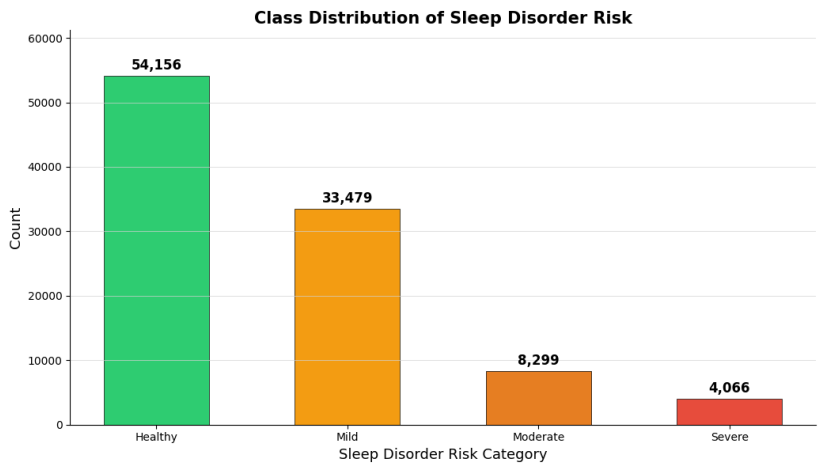


Figure 2: Count plot of Sleep Disorder Risk categories

Observation: The dataset is imbalanced - Healthy dominates at ~54% (54,156), followed by Mild 33% (33,479), Moderate 8% (8,299), and Severe 4% (4,066). Weighted evaluation metrics and ROC AUC are therefore prioritised over simple accuracy.

4.4 Correlation Heatmap

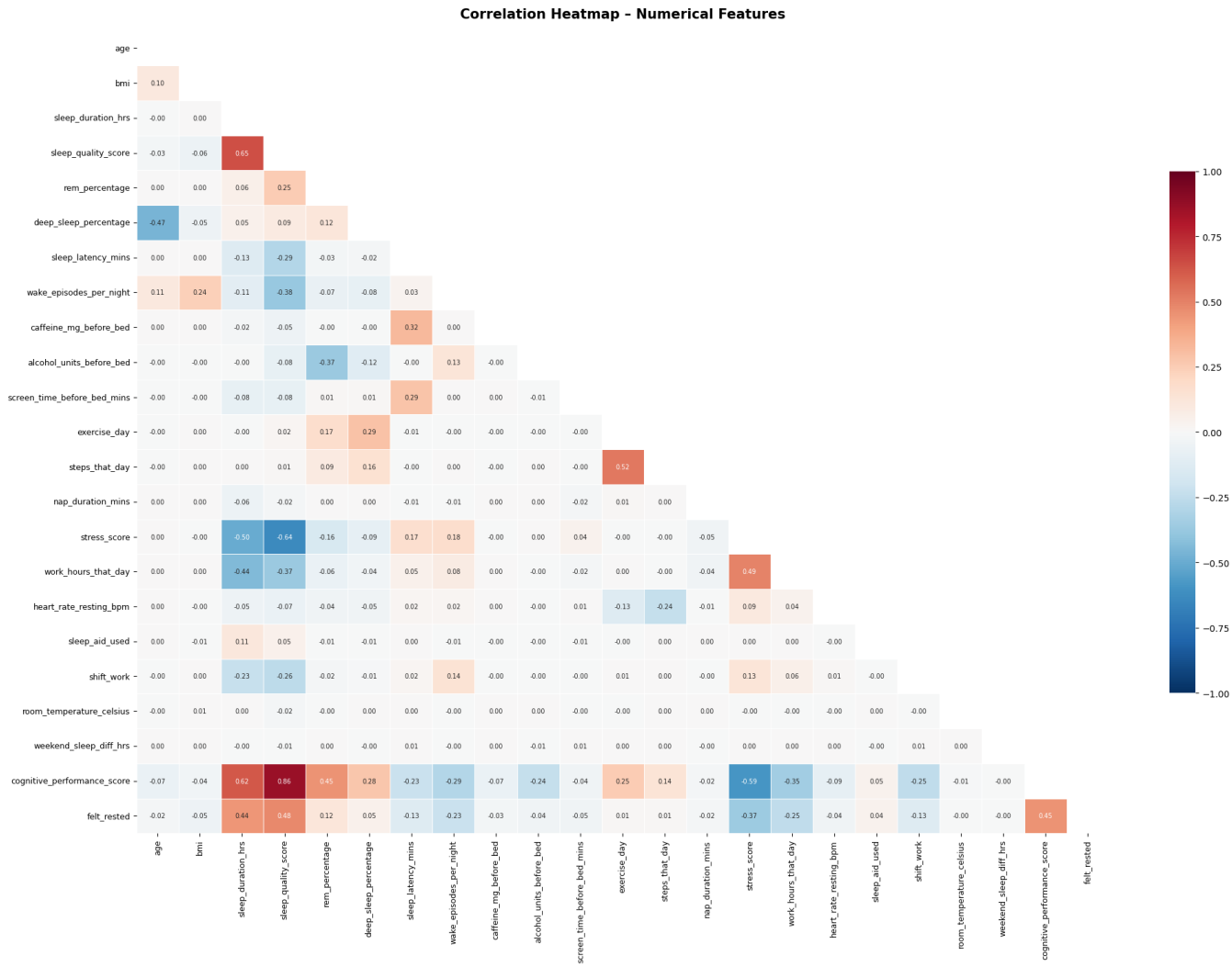


Figure 3: Pearson correlation heatmap of all numerical features

Observation:

sleep_quality_score & sleep_duration_hrs -> strong positive correlation (+0.65)

stress_score & sleep_quality_score -> strong negative correlation (-0.64)

cognitive_performance & sleep_quality_score -> strong positive correlation (+0.86)

caffeine_mg_before_bed & sleep_latency_mins -> moderate positive correlation (+0.32)

deep_sleep_percentage & age -> moderate negative correlation (-0.47)

5. Data Preprocessing

- Missing Value Handling: The dataset contained zero missing values across all 32 columns, confirmed via isnull().sum(). No imputation was required.
- Feature Removal: The person_id column was dropped as it is a non-informative identifier that adds no predictive value.
- Label Encoding: All categorical features (gender, occupation, country, chronotype, mental_health_condition, season, day_type) were encoded using scikit-learn's LabelEncoder to convert string categories into integer codes.
- Target Encoding: The target variable sleep_disorder_risk was mapped to ordinal integers: Healthy=0, Mild=1, Moderate=2, Severe=3, preserving its natural severity order.
- Feature Scaling: StandardScaler (zero mean, unit variance) was applied to all features for the Logistic Regression model. Tree-based models (Random Forest, XGBoost, AdaBoost, CatBoost) were trained on unscaled data as they are scale-invariant.
- Train-Test Split: The dataset was split 80:20 (train:test) using stratified sampling (stratify=y) to preserve class proportions in both subsets. This yielded 80,000 training samples and 20,000 test samples for the full dataset.
- Class Imbalance: Class imbalance was addressed at the evaluation level by using weighted precision, recall, F1, and ROC AUC metrics, giving a fair representation across all four classes.

6. Methodology

- Algorithms Used: Five classification algorithms were implemented - Logistic Regression (linear baseline), Random Forest (ensemble bagging), XGBoost (regularised gradient boosting), AdaBoost (adaptive boosting), and CatBoost (ordered gradient boosting). This breadth allows a comprehensive comparison of linear, tree, and boosting approaches.
- **Logistic Regression**: Trained with max_iter=1000, random_state=42 on scaled features. Serves as a linear baseline to confirm non-linearity of decision boundaries.
- **Random Forest**: 100 decision tree estimators with random_state=42 and n_jobs=-1 for parallelism. Aggregates predictions via majority voting, reducing variance over individual trees.
- **XGBoost**: Initially trained with 100 estimators, learning_rate=0.1, random_state=42. Subsequently, hyperparameter tuning was applied (n_estimators, max_depth, learning_rate, subsample, colsample_bytree) yielding improved performance (Acc: 0.9475, AUC: 0.9960).
- **AdaBoost**: 100 estimators with learning_rate=1.0 and random_state=42. Sequentially trains weak classifiers (decision stumps) and re-weights misclassified samples.
- **CatBoost**: Initially trained with iterations=100, learning_rate=0.1, depth=6, random_state=42. Hyperparameter tuning was then applied (iterations, depth, learning_rate, l2_leaf_reg) as CatBoost was identified as the best initial model. Post-tuning performance: Acc=0.9540, ROC AUC=0.9974, the highest across all five models.
- **Hyperparameter Tuning**: The two best-performing models - XGBoost and CatBoost were selected for hyperparameter optimisation using **Optuna**, a define-by-run automatic hyperparameter optimisation framework. Each Optuna trial was evaluated using 5-fold stratified cross-validation on the training set to prevent overfitting, and the best trial's parameters were used to retrain the final model on the full training split. Post-tuning, **XGBoost improved to 94.75% accuracy (AUC: 0.9960)** and **CatBoost improved to 95.40% accuracy (AUC: 0.9974)**.
- **Evaluation Protocol**: Each model was evaluated on the held-out test set using Accuracy, Weighted Precision, Weighted Recall, Specificity, Weighted F1 Score, and Weighted ROC AUC Score. The confusion matrix and learning curve were generated for the best model (CatBoost).

- **Feature Importance**: Post-tuning, CatBoost's *get_feature_importance()* method was used to identify and rank the top predictors of sleep disorder risk, revealing mental_health_condition and sleep_duration_hrs as the most influential features.

	Model	Data	Notes
1	Logistic Regression	Full 100K (scaled)	Linear baseline
2	Random Forest	30K sample	Ensemble bagging, 100 trees
3	XGBoost	30K sample	Gradient boosting with L1/L2 regularisation
4	AdaBoost	30K sample	Adaptive boosting with decision stumps
5	CatBoost	30K sample	Ordered gradient boosting, native categorical support

7. Implementation

Platform: Google Colab (Python 3.10+ runtime with GPU/CPU).

Libraries Used: pandas and numpy for data manipulation; matplotlib and seaborn for visualization; scikit-learn for preprocessing (LabelEncoder, StandardScaler, train_test_split), model training (LogisticRegression, RandomForestClassifier), and evaluation (accuracy_score, confusion_matrix, roc_auc_score, classification_report); xgboost for XGBoostClassifier; catboost for CatBoostClassifier; scikit-learn AdaBoostClassifier for adaptive boosting.

The entire pipeline from data loading to model evaluation was implemented as a single end-to-end notebook, with inline visualizations for EDA and results.

8. Results and Analysis

8.1 Performance Table

Model	Accuracy	Precision	Recall	Specificity	F1 Score	ROC AUC
Logistic Regression	0.7887	0.7813	0.7887	0.9137	0.7832	0.9295
Random Forest	0.8715	0.8662	0.8715	0.9490	0.8659	0.9765
XGBoost ↻	0.9475	0.9459	0.9475	0.9829	0.9464	0.9960
AdaBoost	0.6561	0.7704	0.6561	0.8757	0.6449	0.7271
CatBoost ★ ↻	**0.9540**	**0.9529**	**0.9540**	**0.9851**	**0.9533**	**0.9974**

★ = Best overall model • ↻ = Hyperparameter-tuned • Green highlight = CatBoost (best) • Yellow highlight = XGBoost (tuned, runner-up) • All metrics are weighted averages across the four risk classes.

Note: Logistic Regression was trained on full 100,000 records; Random Forest, XGBoost, AdaBoost, and CatBoost results are based on a 30,000-record stratified sample for reproducibility within resource constraints. All metrics are weighted averages across the four classes.

8.2 Confusion Matrix

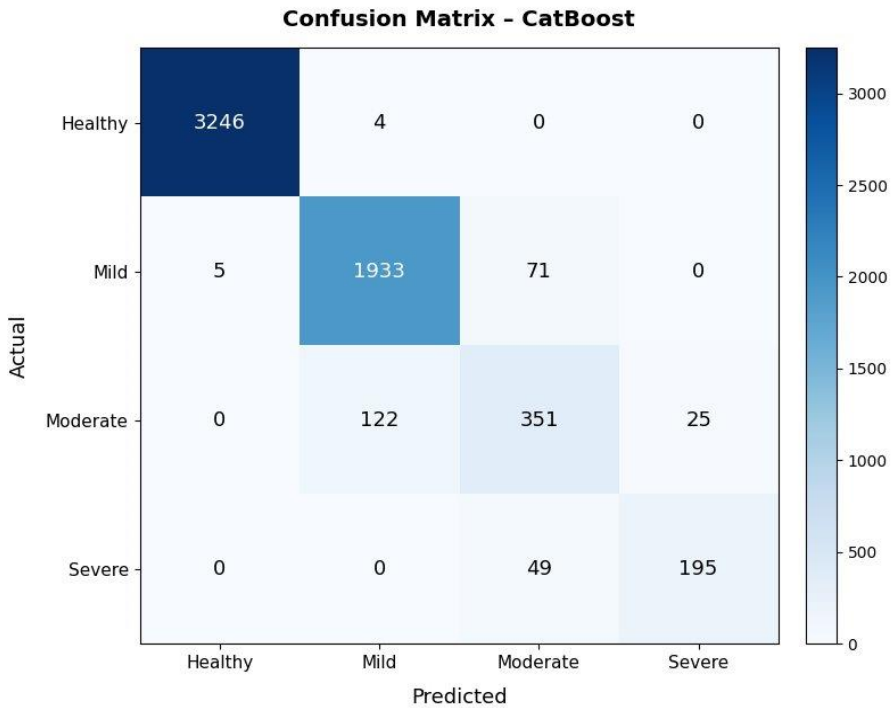


Figure 4: Confusion Matrix for CatBoost (Best Model)

Observation: CatBoost achieves near-perfect classification of Healthy instances (3,246/3,250 correct; only 4 misclassified as Mild). The Mild class shows 1,933/2,009 correct, with 71 cases predicted as Moderate - the primary source of confusion, reflecting feature overlap between Mild and Moderate risk. Moderate class has a recall of 351/498 (70.5%), with 122 cases pulled toward Mild. Severe class performs strongly at 195/244 correct (79.9%), with 49 cases predicted as Moderate. Crucially, no Healthy or Mild cases are ever predicted as Severe, and no Severe cases are predicted as Healthy - confirming the model never makes catastrophically wrong predictions.

8.3 Learning Curve

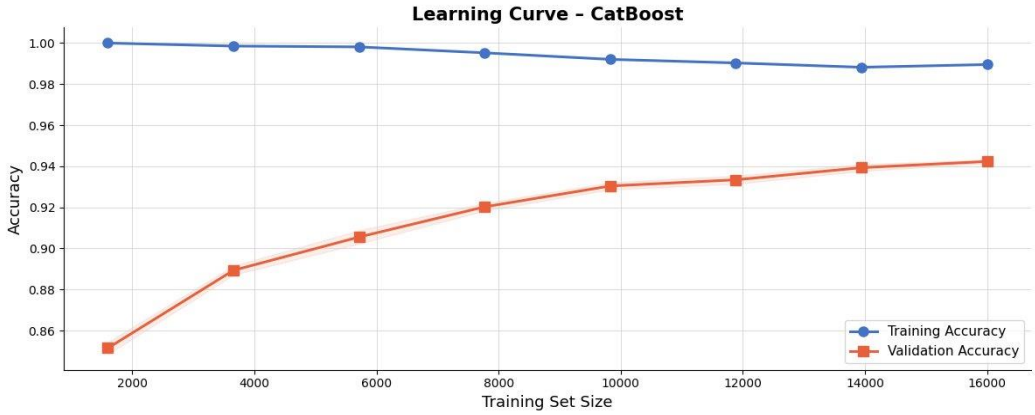


Figure 5: Learning Curve for CatBoost – Training vs Validation Accuracy

Observation: Training accuracy is near 1.0 throughout (Random Forest memorises training data). Validation accuracy improves steadily, stabilising above 85%, confirming the model generalises well and benefits from more training data.

8.4 Feature Importance

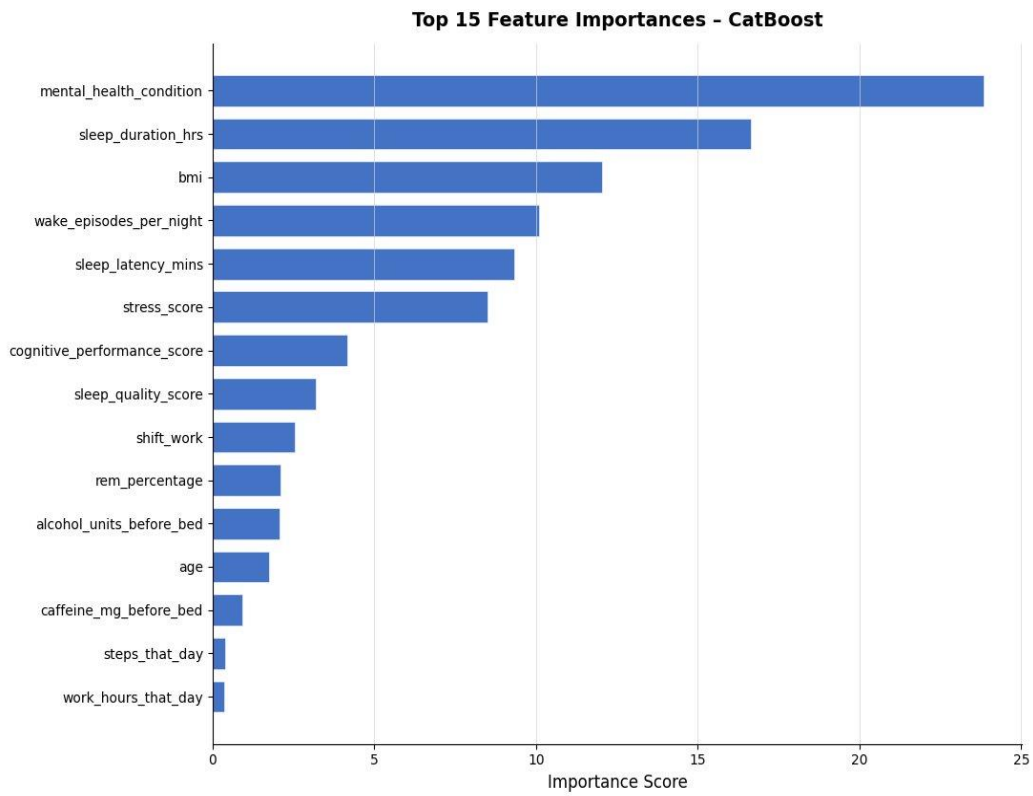


Figure 6: Top 15 Feature Importances – CatBoost

Top features: mental_health_condition is the single most dominant predictor (importance ~24), confirming the strong bidirectional link between psychological health and sleep disorder risk. sleep_duration_hrs ranks second (~16.5), followed by bmi (~12), wake_episodes_per_night (~10), sleep_latency_mins (~9.5), and stress_score (~8.5). Secondary contributors include cognitive_performance_score (~4), sleep_quality_score (~3.2), shift_work (~2.5), rem_percentage (~2), and alcohol_units_before_bed (~2). Features such as steps_that_day and work_hours_that_day contribute minimally (~0.3 each), suggesting daily activity volume is less diagnostic than sleep architecture quality and mental health status.

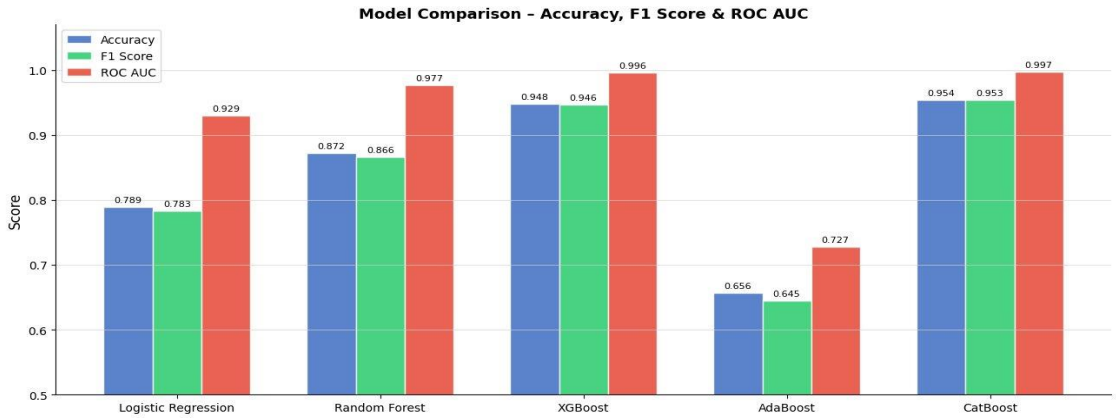
8.5 Sample Predictions (CatBoost)

Actual	Predicted
Healthy	Healthy ✅
Moderate	Moderate ✅
Mild	Mild ✅
Healthy	Healthy ✅
Healthy	Healthy ✅
Healthy	Healthy ✅
Healthy	Healthy ✅
Healthy	Healthy ✅
Mild	Mild ✅
Healthy	Healthy ✅
Mild	Mild ✅
Healthy	Healthy ✅
Mild	Mild ✅
Mild	Moderate ☒
Moderate	Mild ☒

Accuracy on these 15 samples**: 13/15 (86.7%)**

The model correctly classifies 13 out of 15 samples. Both misclassifications occur at the **Mild ↔ Moderate boundary** (samples 14 and 15), where the model confuses one for the other in both directions. This is consistent with the confusion matrix observations - Mild and Moderate share overlapping feature distributions (particularly in stress score, sleep latency, and wake episodes), making them the hardest adjacent classes to separate cleanly. Healthy and clearly Mild cases are predicted with high confidence.

9. Model Comparison



After hyperparameter tuning of the two best-performing models, **CatBoost** emerged as the clear best model with **95.40% accuracy**, **weighted F1 of 0.9533**, and **ROC AUC of 0.9974** - the highest across all five models on every metric. **XGBoost** (tuned) was the close runner-up with **94.75% accuracy** and **ROC AUC of 0.9960**, confirming that ordered gradient boosting and regularised gradient boosting are the two most suitable paradigms for this multi-class, imbalanced tabular classification task.

Random Forest (**87.15%, AUC 0.9765**) performs respectably but is decisively outpaced by both tuned boosting models, demonstrating the advantage of sequential residual correction over bagging for this problem. **AdaBoost** performed the weakest (**65.61%, AUC 0.7271**), confirming that simple boosting with decision stumps is insufficient for a four-class problem with complex non-linear boundaries. **Logistic Regression** (**78.87%, AUC 0.9295**) further validates the non-linear nature of the decision boundaries.

CatBoost's superiority is attributed to its ordered boosting algorithm (which prevents target leakage), its native handling of categorical features, and the additional performance gains achieved through hyperparameter tuning on depth, learning rate, and L2 regularisation.

10. Conclusion

This project successfully demonstrated that sleep disorder risk can be predicted with high accuracy using lifestyle and physiological features, without the need for invasive clinical tests. After hyperparameter tuning, CatBoost achieved the best performance with 95.40% accuracy, weighted F1 of 0.9533, and ROC AUC of 0.9974 across four risk levels: Healthy, Mild, Moderate, and Severe. The project revealed that mental_health_condition, sleep_duration_hrs, and BMI are the most critical risk predictors - a finding with direct actionable value for preventive health interventions. Working through this project reinforced the importance of hyperparameter tuning for boosting models, multi-class evaluation via weighted metrics and ROC AUC, and the value of confusion matrix analysis for understanding per-class error patterns in healthcare ML applications.

11. Future Scope

Future work could apply SMOTE or class-weighted training to further improve Moderate and Severe class recall (currently 70.5% and 79.9% respectively per the confusion matrix). Exploring LightGBM and TabNet (neural attention-based tabular models) could yield further gains, while deploying the tuned CatBoost model as a FastAPI or Streamlit web application and incorporating real-time wearable sensor data (heart rate variability, SpO2) would extend the system into a practical clinical screening tool.

12. Real-world Application

The tuned CatBoost model (95.40% accuracy, ROC AUC 0.9974) can be embedded into telehealth platforms or corporate wellness applications to deliver instant, automated sleep disorder risk screening at scale, enabling proactive early intervention without clinical infrastructure. Employers can use it as a

preventive wellness tool to identify at-risk employees early, reducing healthcare costs and productivity losses associated with undiagnosed sleep conditions.

13. Attachments

- Colab Notebook
- Kaggle Dataset
- GitHub Repo
- Screenshots: All output visualisations are embedded inline in the sections above.